

Supplementary Materials: Pushing the Physical Limits: Extreme Quadrupedal Fall Recovery Under Multi-Joint Failures

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1 Hyperparameter Ablation for the MLP Architecture

To ensure a rigorous and fair architectural comparison in the main text, we conducted exhaustive hyperparameter sweeps to identify the optimal variant of the encoder-based policy. Specifically, we ablated the temporal capacity (history buffer length) and the topological depth (number of hidden layers) to evaluate their impact on recovery performance across extreme failure topologies.

1.1 Success Rate across Spatial Topologies

Figure 1 illustrates the zero-shot recovery success rates of three representative encoder variants across diverse failure modes (Pure Soft, Pure Lock, and Mixed) and severity levels ($N \in [1, 3]$). The three variants evaluated are: **4length_2layer**: A shallower network with 2 hidden layers and a 4-frame history buffer. **8length_3layer**: A deeper network with 3 hidden layers and a longer 8-frame history buffer. **4length_3layer (opt)**: Optimal network with 3 hidden layers and a compact 4-frame history buffer.

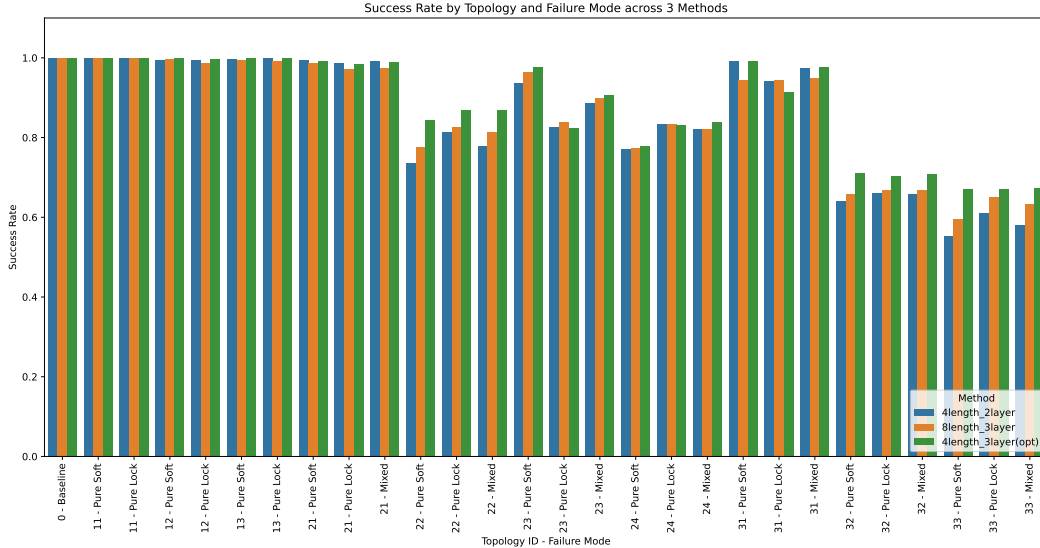


Figure 1: Success Rate by Topology and Failure Mode across 3 Variants. The optimal configuration (4length_3layer) consistently outperforms the variants with longer history (8length) or shallower depth (2layer), particularly in extreme $N = 3$ scenarios.

1.2 Training Convergence Analysis

To further analyze the learning dynamics, Figure 2 presents the training curves of the distinct architecture configurations.

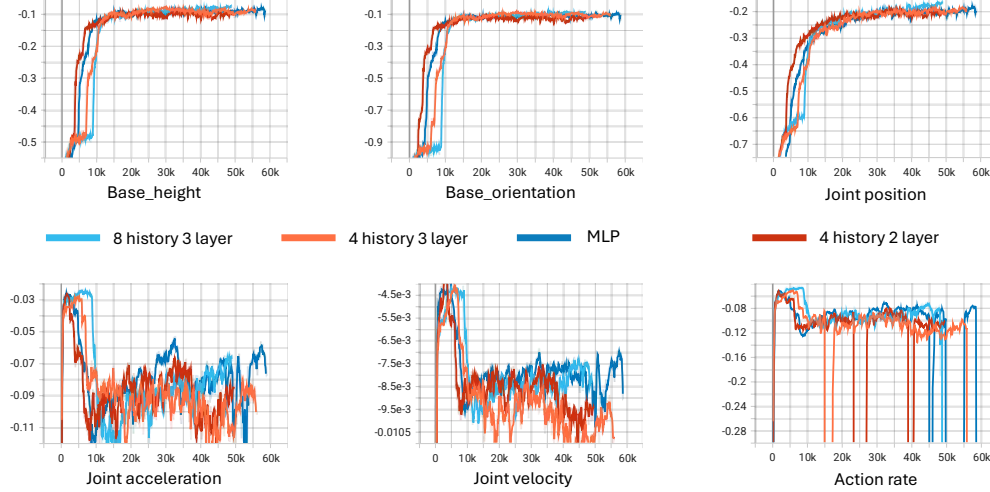


Figure 2: Reward Convergence Curves during Training. The curves illustrate the episodic components of the reward function across the evaluated configurations. All structures can converge.

These extensive ablation studies systematically justify our architectural design choices, confirming that the *4length_3layer* encoder serves as the most robust and reactive backbone for extreme multi-joint failure recovery.